

Fourth-year aerospace engineering student at Kyoto University, researching mission design, trajectory optimization, and robotics.

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Education

2027.04	SOKENDAI · Planned enrollment Five-year doctoral program, Space and Astronautical Science. Graduate Institute for Advanced Studies. Naoya Ozaki's Mission Design Lab.
2023.04 - 2027.03	Kyoto University Fourth-year undergraduate in Engineering Science (Aerospace Engineering). Graduation expected March 2027.
2020 - 2023	Seiko Gakuin High School Kanagawa, Japan
2018 - 2020	Pathways School Gurgaon IB Middle Years Programme · Haryana, India
2017 - 2018	Seiko Gakuin Junior High School Kanagawa, Japan

Research & Development

2025.08 -	Outer-planet mission design and trajectory optimization JAXA / ISAS / Research intern · Naoya Ozaki Lab. Multi-point observations with small outer-planet probes, and low-thrust trajectory optimization with Julia and DDP.
2026.04 -	Model predictive control with a Transformer KYOTO UNIVERSITY / Undergraduate researcher · Kenji Fujimoto Lab. A Transformer-based dynamics model integrated with nonlinear MPC for real-time autonomous tracking on an RC car.
2025.10 -	Reinforcement learning and control for lunar robots KYOTO UNIVERSITY / Research assistant · Learning Machines Group. Reinforcement learning, inverse kinematics, and sim-to-real environments for modular multi-legged lunar robots.
2025.02 - 2025.04	7-DOF robot arm control DFKI · BREMEN / Visiting research fellow · Robotics Innovation Center. Simulation environments and impedance / position controllers for a 7-DOF arm, with space-debris capture as the research context.
2025 -	Athletics record management web app KYOTO UNIVERSITY · TRACK & FIELD / Lead developer & system administrator. A custom React and Firebase application for the Kyoto University Medical School Track & Field Team.
2025.03 -	Russian grammatical error correction TLEEZ INC. / Software engineer intern. AI-based grammatical error correction and training data generation for Russian language learning.
2024.08 -	Metagenomic analysis of microbial communities GEORGIA TECH / Visiting research fellow · DiChristina Lab. Metagenomic analysis of microbial communities involved in heavy metal contaminant stabilization.
2020.12 - 2023.03	Isolation and identification of halophilic archaea TOKYO TECH · ELSI / Research intern · Kosuke Fujishima Lab. Isolation, identification, and phylogenetic analysis of halophilic archaea in rock salt from around the world.

Earlier Research

2023.09 - 2024.08	Microbial metabolism · Kyoto University Wet-lab research on hydrogen and oxygen utilization at the Atomi Lab.
2021.08 - 2022.02	Meteorite analysis · Seiko Gakuin SEM-based composition analysis with UK schools and university researchers.
2020.08 - 2021.08	Artificial gravity · Seiko Gakuin Prototyped a mechanism and built Arduino-based force sensors for evaluation.
2021	Intelligent water rocket A rocket with computer-controlled functions and automated parachute deployment.

Presentations & Papers

2026.08	HDDP-based trajectory optimization for low-thrust GTO-to-GEO transfers 70th UKAREN. Paper submitted.
2026.03	Mission design for multi-point simultaneous observations by ultra-small outer-planet probes on large overseas missions SPS 2026. Oral presentation.
2023.11	Space medicine education in Japan 69th JSAEM. Oral & poster.
2022.06	Extreme halophiles from global rock salts NASA × AGI × High School Students. Online presentation.
2022.05	Isolation and identification of extreme halophiles from global rock salts JpGU 2022. Poster · Award.
2022.03	Research on extreme halophiles Japanese Society of Scientific Fisheries. Online presentation.
2022.01	Meteorite composition analysis with electron microscopy International Research for School. Online presentation.

Awards & Scholarships

2025	Startup Weekend Space Osaka · 2nd place Team leader; business plan and prototype in 54 hours.
2024-25	Nakatani Foundation Scholarship Fully funded research at Georgia Tech and DFKI.
2022	JpGU Award for Academic Excellence Research on halophilic archaea.
2022	Japanese Society of Scientific Fisheries Award Extreme halophiles research.
2022	SDGs Tankyu Award · Nihon Ryoko Prize Founded and organized Seiko SDGs Food Ambassador.
2021	Intelligent Water Rocket · Missions Award Completed all competition missions.
2021	SSH Research Symposium Award Artificial gravity research.
2021	Yurikamome Relay Marathon · Overall winner Team competition.
2019	ISSO National Games · Three golds 1st in 800m, 1500m, and 3000m in India.

Activities

2026.09.15	OpenAI Codex Student Builder Fest 2026 Developed WalksFun, a map service for finding the shortest route through shade, in a three-person team at OpenAI's student hackathon. Served as project manager.
2026.03	JENESYS2025 · Japan-Korea exchange Selected delegate for an international cultural exchange program.
2025.05	Interpreting at the Kyoto International Music Students Festival English-Japanese interpreter and liaison at the Kyoto International Music Students Festival.
2023 -	Mathematics instructor at Tetsuryokukai Advanced mathematics instructor at Tetsuryokukai.
2021	Japan Students Space Forum Founded and organized a forum with more than 100 participants.
2021	Seiko SDGs Food Ambassador Founded a project supporting regional producers through a school festival.
School years	The "Yokohama Banana" project Founded and managed a venture that sold 3,800 chocolate bananas over two days.
2025 / 2023	Space medicine & Moon Village Staff at the 71st JSAEM and 7th Global Moon Village Workshop; organized a university space medicine study tour.
2012 -	Rice farming & Shorinji Kempo Long-term participation in organic rice farming; competed at the national junior-high Shorinji Kempo championship.
2021 / 2019	Science & international exchange Founded the Science Club Summit; organized India-Japan student discussions and a Japanese calligraphy event.
2021	Legislative internship ICAS program at the Tokyo Metropolitan Assembly.

Skills & Qualifications

Programming Python · Julia · C++ · MATLAB · R · Go

Control & robotics DDP · NMPC · Drake · Pinocchio · MuJoCo · Isaac Lab · CasADi · ROS2

Machine learning & web development PyTorch · React · Next.js · Firebase · Docker · Linux · Git

Life science PCR · DNA extraction · PhyML · Metagenomics

Languages Japanese (native) · English (TOEFL iBT 100 / EIKEN Grade 1) · Russian (learning)

Qualifications Model Rocket License Grade 4 · NAUI Open Water Diver · Driver's license (AT)